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A. Notation and Basic Operators

This appendix fixes notation and frequently used operators. Its role is to centralize symbols and conventions so that the subsequent appendices can state routing, computation, and composition claims without repeatedly reintroducing definitions. We work with a compact input domain $X \subset \mathbb{R}^d$ and the prompt-conditioned map F_{θ^*} with hypothesis class H_{θ^*} as introduced in the main text.

Basic symbols. Scalars are denoted by lowercase letters (e.g., $\varepsilon, \tau, \Delta$). Vectors are bold lowercase (e.g., $\mathbf{q}$) and matrices are bold uppercase (e.g., $\mathbf{W}$). For a vector $\mathbf{u}$, $\mathbf{u}_i$ denotes its i-th coordinate; for a matrix $\mathbf{M}$, $\mathbf{M}_{ij}$ denotes its (i, j) entry.

Domain, function space, and norms. Inputs are $x \in X \subset \mathbb{R}^d$ (compact; see Assumption 3.1), and the target function is $g \in C(X)$. For $f : X \to \mathbb{R}$, the sup norm is

$$\|f\|_\infty \;:=\; \sup_{x \in X} |f(x)|.$$

For vectors, $\|\cdot\|_2$ and $\|\cdot\|_\infty$ denote the Euclidean and ℓ_∞ norms, respectively.

Fixed backbone and prompt-conditioning. The fixed executor parameters are θ^*. Prompts are soft prefixes of length-L in the space $\mathcal{P} = \mathbb{R}^{L \times d_p}$ as in Definition 3.2, and we write $p = (p_1, \ldots, p_L) \in \mathcal{P}$. The prompt-conditioned scalar output is $F_{\theta^*}(p, x)$, and the induced prompt-switching family is $H_{\theta^*} = \{F_{\theta^*}(p, \cdot) : p \in P\} \subset C(X)$ as in Definition 3.4.

Error budgets. We use $\varepsilon > 0$ for an end-to-end uniform error target. When splitting the budget, we write $\varepsilon_{\text{approx}}$ for the approximation error (e.g., $g \approx N_g$) and $\varepsilon_{\text{exec}}$ for the executor emulation error, with $\varepsilon_{\text{approx}} + \varepsilon_{\text{exec}} \le \varepsilon$. Local budgets are denoted by δ or η.

Row-wise softmax and attention operators. Let S_τ denote the row-wise softmax with temperature $\tau > 0$, i.e., $S_\tau(r) = \mathrm{softmax}(r/\tau)$ applied to each row. Let the model width be D, and let $W_Q, W_K, W_V \in \mathbb{R}^{D \times D}$. For a token/state matrix $Z \in \mathbb{R}^{n \times D}$, define single-head self-attention by

$$A_\tau(Z) \;:=\; S_\tau\left(\frac{(ZW_Q)(ZW_K)^\top}{\sqrt{D}}\right)(ZW_V).$$

Prompt read as memory access. When prompts are treated as tokens in $\mathbb{R}^D$ (typically by taking $d_p = D$ or by an implicit linear embedding), we write the prompt matrix as $\mathbf{P} = [p_1; \ldots; p_L] \in \mathbb{R}^{L \times D}$ and define

$$K_{\mathbf{P}} := \mathbf{P}W_K, \qquad V_{\mathbf{P}} := \mathbf{P}W_V.$$

Given a query $q \in \mathbb{R}^D$, the prompt read operation is

$$\mathcal{R}_\tau(q; \mathbf{P}) \;:=\; S_\tau(qK_P^\top)\, V_P \in \mathbb{R}^D.$$

Routing analyses use a score margin $\Delta > 0$ and an impurity level $\rho \in (0, 1)$ ("$1 - \rho$ nearly one-hot").

ReLU and FFN. The ReLU is $\sigma(u) = \max\{u, 0\}$. A token-wise two-layer ReLU FFN with hidden width h is

$$\phi(z) \;:=\; W_2\, \sigma(W_1 z + b_1) + b_2, \qquad W_1 \in \mathbb{R}^{h \times D},\; W_2 \in \mathbb{R}^{D \times h},$$

applied row-wise to token matrices (by applying ϕ to each row).

A.1. Symbol table

Table 1. Core notation used throughout the paper.

Symbol	Meaning
$X \subset \mathbb{R}^d$	Compact input domain.
d	Input dimension.
$x \in X$	Input point.
$C(X)$	Continuous real-valued functions on X.
$\|f\|_\infty$	Sup norm $\sup_{x\in X} \|f(x)\|$.
θ^*	Fixed backbone/executor parameters.
$\mathcal{P} = \mathbb{R}^{L\times d_p}$	Soft-prompt space; L prompt length, d_p prompt width.
$p = (p_1, \ldots, p_L) \in P$	A prompt (soft prefix).
$F_{\theta^*}(p, x)$	Prompt-conditioned scalar output.
H_{θ^*}	Prompt-induced family $\{F_{\theta^*}(p, \cdot) : p \in P\} \subset C(X)$.
$g \in C(X)$	Target function.
N_g	ReLU MLP approximator of g (classical UAT step).
ε	End-to-end uniform error target.
$\varepsilon_{\text{approx}}, \varepsilon_{\text{exec}}$	Approximation versus executor error budgets.
δ, η	Local/intermediate error budgets.
D	Model width (token/state dimension).
$Z \in \mathbb{R}^{n\times D}$	Token/state matrix; n is number of tokens.
S_τ	Row-wise softmax with temperature $\tau > 0$.
τ	Temperature parameter (smaller τ sharpens routing).
W_Q, W_K, W_V	Attention projection matrices.
$A_\tau(Z)$	Single-head self-attention operator.
$\mathbf{P} = [p_1; \ldots; p_L] \in \mathbb{R}^{L\times D}$	Prompt viewed in model space (when $d_p = D$ or embedded).
$K_{\mathbf{P}} = \mathbf{P}W_K,\ V_{\mathbf{P}} = \mathbf{P}W_V$	Prompt keys/values.
$q \in \mathbb{R}^D$	Query vector.
$\mathcal{R}_\tau(q; \mathbf{P})$	Prompt read $S_\tau(qK_P^\top)V_P$.
Δ, ρ	Routing margin and impurity level.
$\sigma(\cdot)$	ReLU activation.
$\phi(\cdot)$	Token-wise two-layer ReLU FFN; h is hidden width.

B. Simplified Transformer Architecture

This appendix specifies the simplified Transformer family used throughout the construction. The purpose is to pin down the minimal architectural components needed by the executor: row-wise softmax S_τ, single-head self-attention A_τ, the prompt read operation $\mathcal{R}_\tau(\cdot; P)$, and a token-wise two-layer ReLU FFN $\phi(\cdot)$ (extended row-wise as $\Phi(\cdot)$), all at a fixed model width D.

Tokens, prompt, and input embedding. We take the prompt width to match the model width, i.e., $d_p = D$, so a prompt is a length-L token sequence $p = (p_1, \ldots, p_L) \in \mathcal{P} = \mathbb{R}^{L\times D}$ with $p_i \in \mathbb{R}^D$. An input $x \in \mathcal{X} \subset \mathbb{R}^d$ is embedded into a single input token

$$u(x) \ := \ E_x x + b_x \in \mathbb{R}^D, \qquad E_x \in \mathbb{R}^{D\times d},\ \ b_x \in \mathbb{R}^D,$$